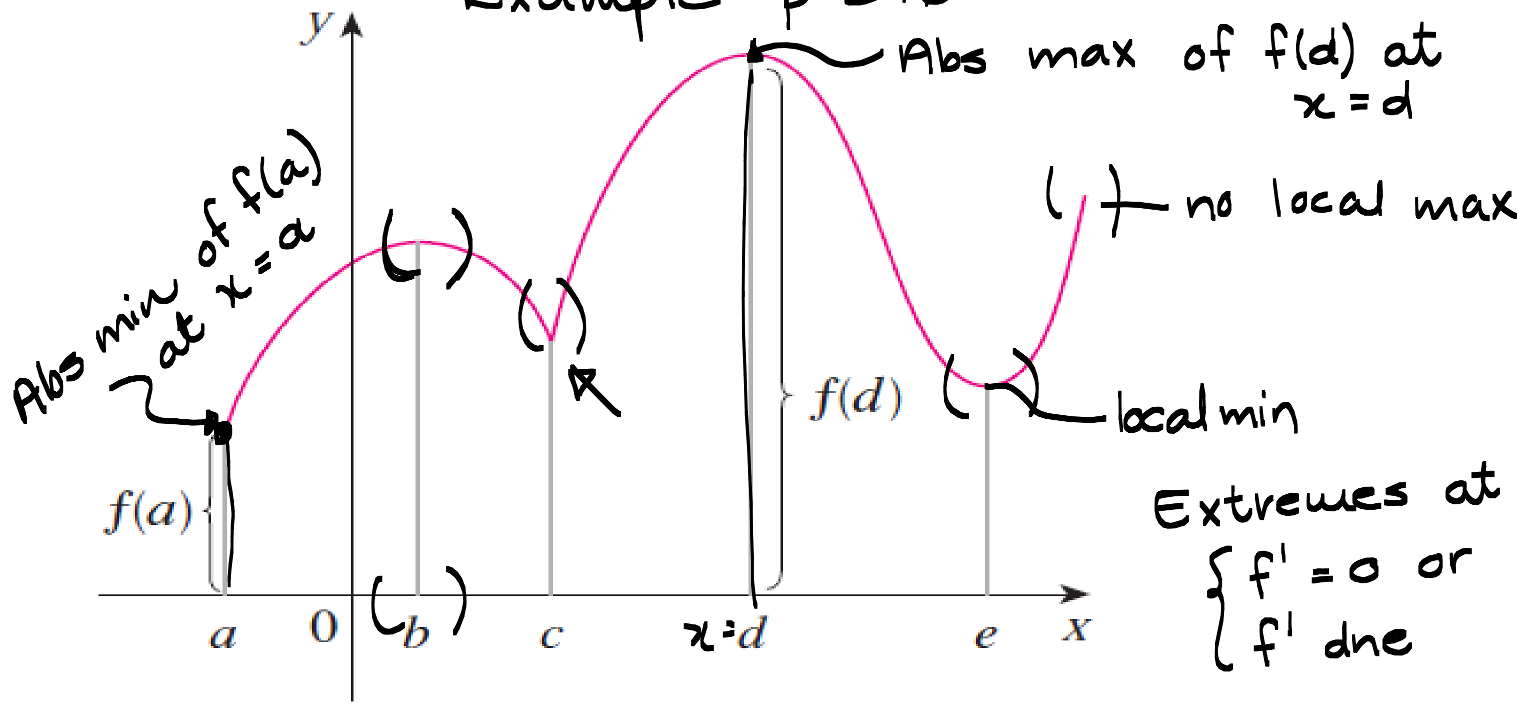
Theme 4: Applications of differentiation.

Unit 4.1 Maximum and minimum values

pp. 276-282

Example p 276



Definition 1: p 276

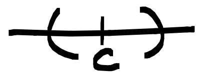
Let c be a number in the domain D of a function. Then f(c) is the

- absolute maximum of f on D if $f(c) \geq f(x) \quad \forall \ x \in D$.
- absolute minimum of f on D if $f(c) \leq f(x) \quad \forall \ x \in D$

Absolute/global extreme values.

Definition 2 The number $f(c)$ is a

- local maximum if $f(c) \geq f(x)$ when x is near c.



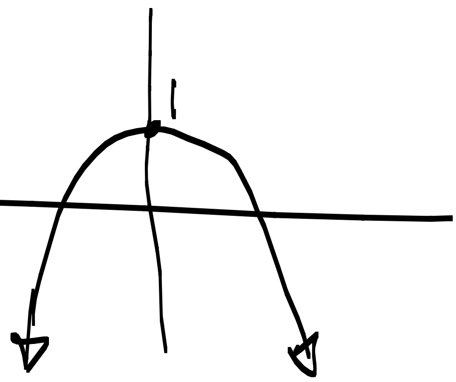
- local minimum if $f(c) \leq f(x)$ when x is near c.

Local max/min — Local extremes.

Example:

Find the extremes of $f(x) = -x^2+1$.

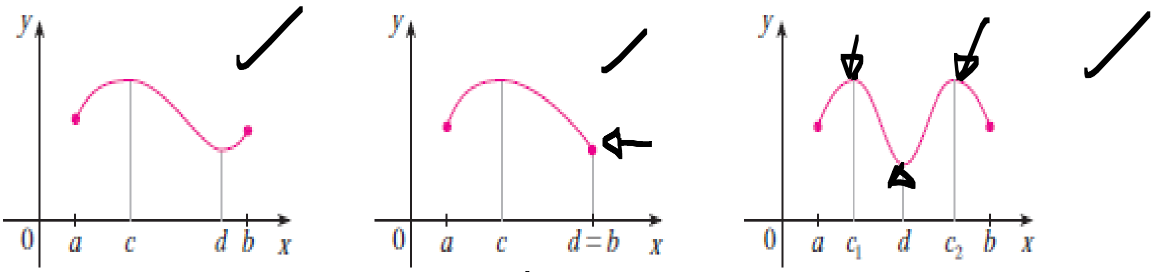
global? local?

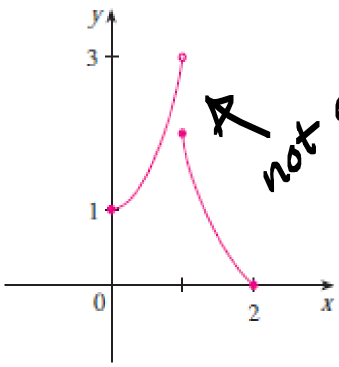


f has an abs. max of $f(0) = 1$ at $x = 0$.

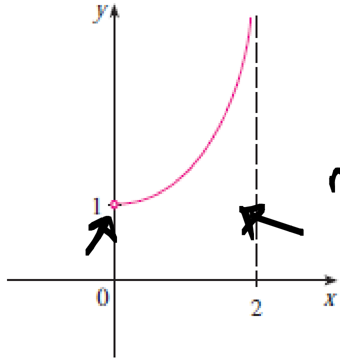
Extreme value theorem p 278

f continuous on closed interval [a,b] then f attains an absolute max value f(c) and an absolute minimum value f(d) at some numbers c and d in [a,b].





not continuous.

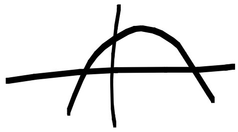


not closed interval.

[0, 2)

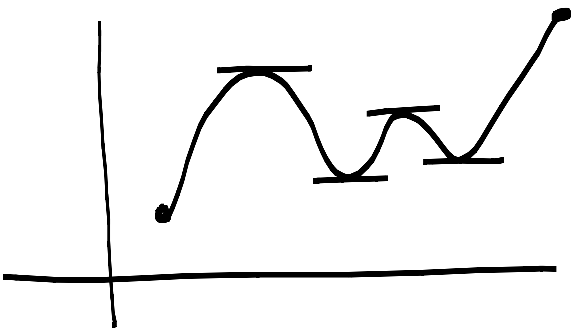
Abs. min of 0 at $x = 2$
No abs. max

No max or min.



Fermat's theorem: p279

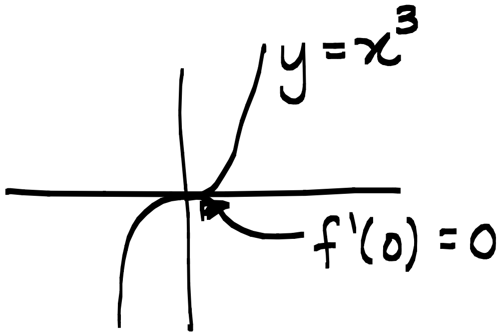
If f has a local max or min at c, and if $f'(c)$ exists, then $f'(c) = 0$.



TP $\Rightarrow$ $f'(c) = 0$

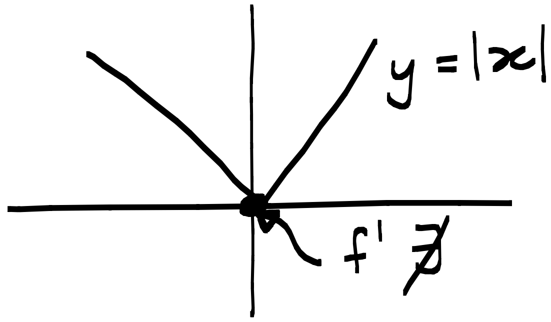
$f'(c) = 0 \Rightarrow$ TP ?

Be careful: The converse of Fermat's theorem is not true.



$f'(0) = 0$

no local max/min



$f'(0)$ does not exist.

yet there is a local min

<u>Definition 6</u> p 280

A critical number of f is a number c in the domain of f such that

$f'(c) = 0$ or $f'(c)$ does not exist.

<u>Example</u> Also see example 7 p 280

① Find the critical values of the function $f(x) = (x-1)\cdot x^{2/3}$

$$f'(x) = x^{2/3} + (x-1)\,\tfrac{2}{3}\,x^{-1/3}$$

f' does not exist if $x = 0$.

$$f'(x) = x^{2/3} + \frac{2(x-1)}{3x^{1/3}} = 0 \qquad \text{(CV's)}$$

$$\Rightarrow 3x + 2(x-1) = 0$$

$$\Rightarrow x = \frac{2}{5}$$

$\Rightarrow$ CV's at $x = 0$ and $x = \frac{2}{5}$

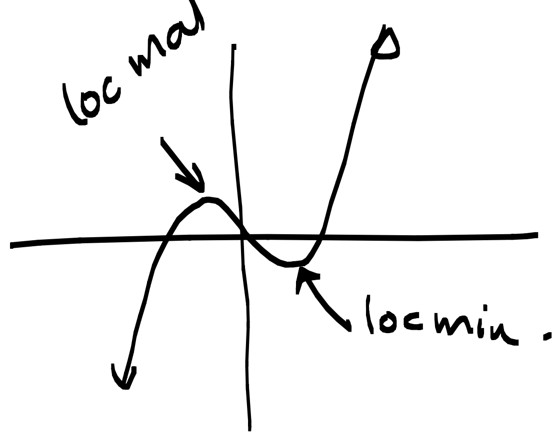
② Find the critical values of the function and use a graph to decide whether the function has extreme values.

1. $f(x) = x^3 - 3x$

$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

and f' exists everywhere.

$\therefore$ CV's $x = \pm 1$



No global extreme.

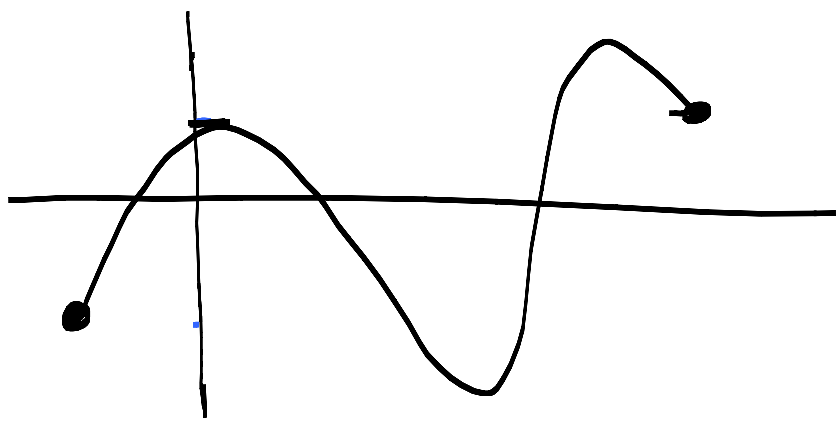
Local max at $x = -1$ of $f(-1) = 2$

$\Rightarrow$ local min at $x = 1$ of $f(1) = -2$.

To find an absolute max or min of a continuous function on a closed interval:

The closed interval method p 281

1. Find the critical values of f in (a,b)
2. Find the function values of f at $\boxed{\text{critical points}}$ and $\boxed{\text{endpoints.}}$
3. Largest – absolute maximum.
   Smallest – absolute minimum.



Example: Find the extremes of

① $f(x) = x^3 - 3x^2 + 1$, $x \in [-\frac{1}{2}, 4]$

1. $f'(x) = 3x^2 - 6x = 0$

$\Rightarrow x(x-2) = 0 \Rightarrow x = 0$ or $x = 2$

$f'(x)$ exists for all $x \in [-\frac{1}{2}, 4]$

So CV's $x = 0$ and $x = 2$

2. For extreme values:

CV's $\begin{cases} f(0) = 1 \\ f(2) = -3 \end{cases}$

Endpoints $\begin{cases} f(-\frac{1}{2}) = \frac{1}{8} \\ f(4) = 17 \end{cases}$

3. $f(4) = 17$ is the global max at $x = 4$.
$f(2) = -3$ is the global min at $x = 2$.

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(2) Find extreme values of $f(x) = x^4 - 2x^2$ on $[-0.5, 2]$.

$f'(x) = 4x^3 - 4x = 0$

$\Rightarrow x(x-1)(x+1) = 0$

$\therefore x = 0, 1, -1$ and

$f'(x)$ exists on $\mathbb{R}$ so CV's on $[-0.5, 2]$ is $x = 0$ and $x = 1$. ($x = -1 \notin [-0.5, 2]$)

$f(0) = 0$ $\quad$ $f(-0.5) = -0.437$

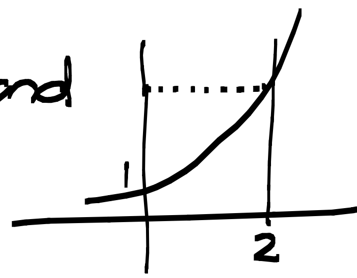
$f(1) = -1$ $\quad$ $f(2) = 8$

Abs. max is 8 at $x = 2$.
Abs. min is -1 at $x = 1$

(3) $f(x) = e^x$, $x \in [0, 2]$

$\therefore f'(x) = e^x > 0$ for all x so

Abs max at $x = 2$, $f(2) = e^2$ and
abs min at $x = 0$, $f(0) = 1$



④ $f(x) = -x^2 + 1, \quad x \in [\frac{1}{2}, 3]$

So $f'(x) = -2x < 0$ for $x \in [\frac{1}{2}, 3]$

$\Rightarrow$ Abs max at $x = \frac{1}{2}$, $f(\frac{1}{2}) = \frac{3}{4}$

and abs min at $x = 3$, $f(3) = -8$